

Two-echelon Vehicle Cooperated Operations Problems: The State of the Art and Future Direction

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Abstract. In the Two-Echelon Vehicle Cooperation Problem (2ECOP), different vehicles operate on primary and secondary echelons to optimize package delivery, resulting in cost savings and efficient services. Our work focuses on the cooperation of two-echelon vehicles and their optimal scheduling and routing strategies, as well as their numerous applications across industry. We examine the literature of the 2ECOP across various dimensions, including various cooperated modes, distinct optimization methodologies, and practical settings. Our work identifies various effective algorithms and heuristic solution procedures for the 2ECOP. We also propose a unified approach for delivery that is designed for reducing cost and improving service efficiency. We believe that academic research and industry would both benefit from a unified classification scheme that extends to new two-echelon operated logistics models that may emerge in response to future technological advances.

Keywords: Two-echelon vehicles, delivery, literature review, future research directions

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1. Introduction

1.1 Background

With the growth of modern information and communication technologies (such as 5G and AI), companies including Google, Alibaba, UPS, DHL, and FedEx are exploring two-echelon delivery methods (Ha et al. 2018). In the Two-Echelon Vehicle Cooperation Problem (2ECOP), First-Echelon (FE) vehicles transport goods from central depots to local depots or directly to some customers, after which Second-Echelon (SE) vehicles distribute the goods to other designated customers. This approach

integrates concepts from the Vehicle Routing Problem (VRP) and the Traveling Salesman Problem (TSP), along with their variants, to achieve cost reduction and improve service efficiency. 2ECOP models are applicable to a wide range of practical challenges in various industries, including transportation, supply chain management, and production planning. For example, JD.com has successfully implemented drone delivery services in several provinces in China. Similarly, Alibaba Group and Baidu have deployed package copters for home delivery (Liu et al. 2021). These examples illustrate the growing importance and di-

verse applications of two-echelon delivery systems in modern logistics.

Convergent technology is advancing rapidly, and drones/robots are being applied in various fields of industry and are commonly used for commercial delivery. Starting with the work of [Murray and Chu \(2015\)](#), the collaborative human-operated truck(s) and drone(s) delivery team has received much attention recently, and there are several works on this collaborative system (see for example, [Chung et al. \(2020\)](#); [Macrina et al. \(2020\)](#); [Moshref-Javadi and Winkenbach \(2021\)](#); [Liang and Luo \(2022\)](#); [Madani and Ndiaye \(2022\)](#)) that provide structured reviews of specific subsets of drone-related planning problems. Given the limited capacity of drones, it is logical to use a ground vehicle or a (possibly automated) robot in tandem to increase capacity, efficiency, and effectiveness. The work of [Boysen et al. \(2018\)](#) is apparently the first to combine scheduling and routing decisions in the context of two-echelon urban logistics for robotic parcel delivery. [Srinivas et al. \(2022\)](#) provide a systematic survey of existing research on autonomous delivery robot-based delivery systems and their associated operations. To ensure alignment with real-world technological advances and operational needs, we focus on cooperative truck and drone/robot operations.

From the perspective of two-echelon systems, [Li et al. \(2021\)](#) and [Sluijk et al. \(2023\)](#) provide insights into two-echelon routing problems. [Li et al. \(2021\)](#) classify these problems by focusing on the modeling mechanisms that connect the two echelons, specifically addressing Two-Echelon Vehicle Routing Problems

(2E-VRP) and Two-Echelon Location Routing Problems (2E-LRP). [Sluijk et al. \(2023\)](#) provide a comprehensive review of a subset of 2ECOP; however, their work does not provide a systematic and scalable classification framework based on vehicle roles that distinguishes between different two-echelon cooperative operations logistics models, particularly in terms of detailed design and operational configurations. Our research builds on these contributions by emphasizing the 2ECOP with a particular focus on delivery modes, deliberately excluding the papers covered in their review to present a unique and non-redundant perspective.

After a thorough review of the literature on two-echelon routing problems (e.g., [Chung et al. \(2020\)](#); [Macrina et al. \(2020\)](#); [Srinivas et al. \(2022\)](#); [Sluijk et al. \(2023\)](#)), we observe that 2ECOP problems can be systematically classified based on the usage scenarios of two types of vehicles. Several studies focus on drone-resupply scenarios where the drone acts as the FE vehicle, while the truck acts as the SE vehicle. For instance, [Dayarian et al. \(2020\)](#) consider a home delivery system where delivery trucks are regularly resupplied by drones. [Pina-Pardo et al. \(2021\)](#) examine a scenario in which a truck follows a delivery route and receives new orders en route through drone resupply from the depot. Similarly, [Dienstknecht et al. \(2022\)](#) investigate a system where the remaining shipments are resupplied to the truck on its route by a single drone. In these scenarios, the drone's primary function is not to perform final deliveries to customers but rather to resupply the delivery vehicle, which then completes the last-mile delivery. Based on whether

the FE vehicle (typically truck(s)) makes deliveries, we propose a classification of 2ECOP into two macro-classes:

- i) The FE vehicle functions as a moving depot while only the SE vehicles perform deliveries. This class includes three scenarios: (a) A tour with a single FE vehicle and local depots, where the logistics provider's vehicle starts at a central depot and visits dedicated locations to pick up and release the SE vehicles within the delivery area; (b) A tour with a single FE vehicle and drop-off points, where the FE vehicle stops at predefined drop-off points for releasing SE vehicles. During the tour, no new SE vehicles can be loaded. The FE vehicle, however, has the flexibility to pick up SE vehicles either at their original drop-off points or at subsequent drop-off points; (c) Multiple tours with several FE vehicles, where each FE vehicle operates its own independent route incorporating both local depots and drop-off points, allowing for improved customer service and reduced delays compared to scenarios (a) and (b). This configuration highlights the flexibility of assigning distinct routes to each FE vehicle to enhance operational efficiency.
- ii) Deliveries may be performed by either the FE vehicle(s) or the SE vehicle(s). This class consists of two scenarios: (d) a tour with a single FE vehicle, and (e) multiple tours with two FE vehicles, which extends scenario (c) to include multiple vehicles for simultaneous delivery to different customers. Fig.1 shows examples of this classification.

1.2 Review Methodology

We limit our scope to papers published in the past six years (i.e., 2018-2023). To examine the relevant literature, we adapt a systematic review methodology consisting of three distinct stages: planning, conducting, and reporting (see [Cai and Choi \(2021\)](#), [Srinivas et al. \(2022\)](#)). In the first step involving planning, the critical task is to identify keywords that retrieve an exhaustive list of research works employing two-echelon vehicle cooperated operations. We search 15 combinations of keywords, as shown in Table 1, which includes components of vehicles (Category A), and applications problems (Category B). The chosen terminology has proven to be consistently suitable across our sample papers, enabling effective collection of the relevant literature. Utilizing secondary keywords, we embark upon an advanced search query using Google, successfully retrieving 200 research articles at the second stage.

To refine the selection, we establish exclusion and inclusion protocols based on Google's guidelines, manually adding more relevant articles while filtering out irrelevant ones. [Imran et al. \(2023\)](#) consider a fully automated system instead of human-machine cooperation, so their content is not included here. Then, we review the titles and abstracts of all other retrieved papers, with a primary focus on excluding studies related to trucks and drones for parallel delivery. Ultimately, 66 papers are deemed suitable for review. In view of the nascent stage of the 2ECOP literature, in addition to reviewing these selected studies, we also study multiple review papers that direct readers to state-of-the-art research in related,

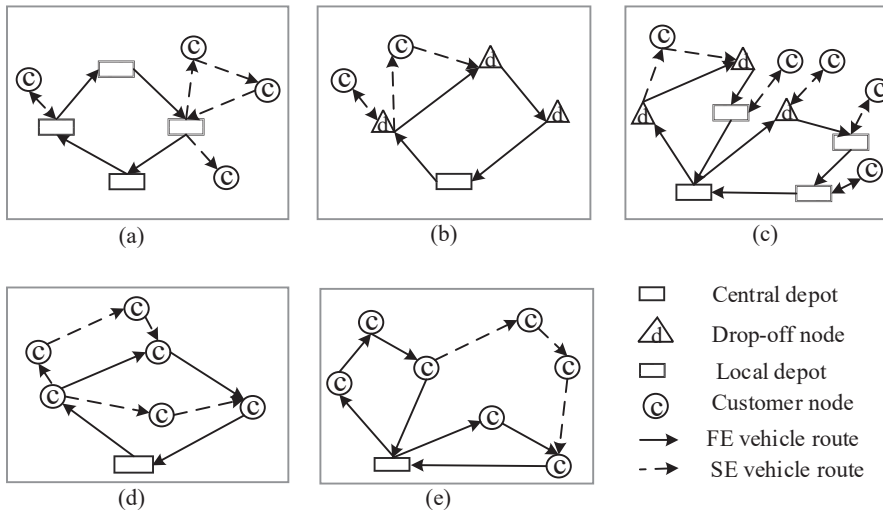


Figure 1 Examples of FE and SE Vehicle Coordinated Delivery Methods

Table 1 Categories of Words Utilized for the Literature Search

Categories A	Categories B
Truck(s); Robot(s); Unmanned Aerial Vehicles (UMAs, such as Drones); Two-echelon vehicle(s)	Vehicle routing problems; Traveling salesman problems; Last-mile delivery problems

Note: Our search process combines keywords in Categories A with keywords in Categories B resulting in $5 \times 3 = 15$ search combinations.

more mature fields, thereby facilitating further research. The final stage of our review methodology involves comprehensive reporting. Fig.2 shows the evolution of these 74 publications over time, with a strong overall increasing trend.

1.3 Contribution and Organization

In this paper, we review different mathematical models, solution methods, trade-off between operating cost and responsiveness. The research questions found in the literature can be summarized as follows:

- 1) How can two-echelon vehicle systems be classified based on their relevant parameters and characteristics?
- 2) What methodologies and optimization techniques can be employed to deter-

mine the optimal scheduling and routing strategies for two-echelon vehicles, considering the inherent complexities and trade-offs between operating costs and responsiveness?

- 3) In what practical scenarios and industries can the concept of two-echelon vehicles be applied effectively, and what are the key considerations for successful implementation?

This literature review is structured as follows. Section 2 formally defines the 2ECOP. In Section 3, we discuss algorithms for the 2ECOP along with various applications explored in the two-echelon literature. Section 4 identifies future research directions. Section 5 contains some concluding remarks.

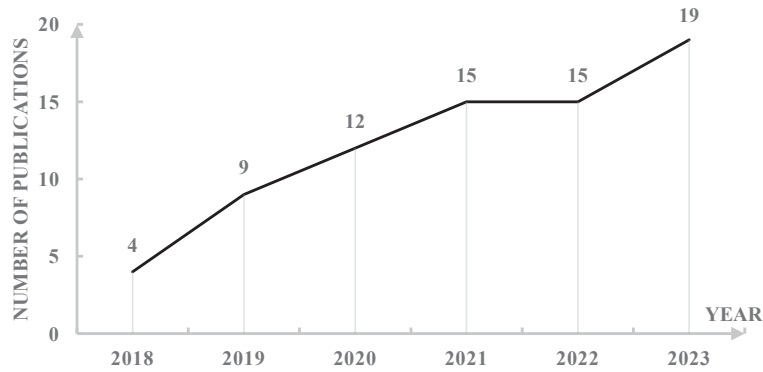


Figure 2 Number of Publications Per Year

2. Two-Echelon Vehicle Cooperated Operations

The two-echelon cooperative operation problem involves coordinating distinct operational units, typically trucks/vans (first-echelon or FE vehicles) and drones/robots (second-echelon or SE vehicles), to optimize logistics tasks. A key aspect of 2ECOP is defining the roles of each vehicle type - determining which vehicles serve as FE units for primary transportation and which operate as SE units for final deliveries. Both vehicle types execute delivery tasks either independently or collaboratively, with the system's configuration and objectives focusing on optimizing the allocation, routing, and synchronization of both echelons to maximize efficiency while meeting constraints such as time windows, vehicle capacities, and energy limitations. Two primary operational modes emerge: SE vehicle-only delivery, where SE vehicles independently handle customer deliveries; and FE and SE vehicle co-deliver, where SE vehicles collaborate with FE vehicles to serve customer demands jointly. These modes highlight distinct approaches to leveraging two-echelon systems, each with unique challenges and advantages. Detailed

discussions of these two macro-classes are presented in Sections 2.1 and 2.2, respectively.

2.1 SE Vehicle-only Delivery

The FE vehicle acts as a mobile depot and recharging platform for the SE vehicles. The SE vehicles may dispatch from the FE vehicle with one or more packages, and deliver these packages to their designated destinations, allowing each customer to be served by an SE vehicle. There are several papers describing SE vehicle delivery exclusively; Table 2 provides a summary of the main features of this delivery design. The table highlights key aspects, including the number of FE and SE vehicles considered (e.g., 1, n , m , or ∞), the objective of the problem, the energy capacity of SE vehicles, the allowance for multiple visits, the presence of time-window constraints for customers, and the classification of delivery types. Here, the delivery class specifically represents the categorization within 2ECOP. We now discuss these papers in detail, focusing on the first category of the 2ECOP classification.

Each SE vehicle operates by delivering to customers and returning to assigned local depots, as explored in studies by Boysen et al. (2018), Alfandari et al. (2022), and Momeni

Table 2 Summary of the Main Features of SE Vehicle Delivery Only

Authors	Number of		Objective	SE energy	SE capacity	SE multiple visits	Time windows	Delivery class
	FE	SE						
Boysen et al. (2018)	1	m	min-weighted lateness			✓		(a)(b)
Chang and Lee (2018)	1	m	min-time			✓		(b)
Poikonen and Golden (2020)	1	m	min-time	✓	✓	✓		(b)
Liu et al. (2021)	n	m	min-cost	✓	✓	✓	✓	(c)
Alfandari et al. (2022)	1	m	min-time			✓	✓	(a)
			min-total					
Dukkanci et al. (2023)	n	m	unsatisfied demand	✓	✓		✓	(c)
Momeni et al. (2023)	n	m	min-time	✓	✓			(a)

et al. (2023). Boysen et al. (2018) investigate a truck-based freight delivery system where a truck transports goods from a central depot to a city center, using decentralized robotic depots for final deliveries. Their study highlights the importance of minimizing late deliveries, with results emphasizing how truck waiting times for returning robots significantly increase customer dissatisfaction. Alfandari et al. (2022) also examine an urban delivery system where a single FE vehicle transports packages from a central depot to designated local facilities. SE devices, such as robots, are deployed from these facilities for last-mile deliveries. By analyzing various tardiness metrics (e.g., maximum tardiness, total tardiness, and the number of late deliveries), their results demonstrate the sensitivity of delivery performance to the number of operational local facilities. Further, Momeni et al. (2023) investigate drone delivery operations supported by multiple trucks acting as mobile depots for charging drone batteries, with the objective of minimizing total delivery time. Their results indicate that re-

duced battery capacity significantly shortens drone flight durations, thereby requiring the deployment of additional local depots to mitigate these limitations.

Drop-off nodes play a critical role in collaborative truck-drone delivery systems, where the FE vehicle releases SE vehicles for further distribution. Chang and Lee (2018) focus on optimizing parcel delivery by having a truck depart from a central depot to visit cluster centers formed by multiple delivery locations. At each cluster center, the truck acts as a drop-off point, dispatching multiple drones simultaneously to deliver parcels within the cluster before returning to the truck. Their findings demonstrate that integrating drones with trucks can significantly improve delivery efficiency. Similarly, Poikonen and Golden (2020) explore a tandem routing problem involving a truck and multiple drones. In their model, the truck serves as a mobile drop-off node, enabling drones to depart with multiple packages, deliver them to designated locations, and return for recharging or additional pick-

ups. Their results emphasize the importance of drop-off nodes in enhancing the flexibility and efficiency of truck-drone delivery systems.

Incorporating multiple FE vehicles enhances the capability of delivery systems to handle larger volumes of packages, especially when supported by local depots or drop-off points. Dukkanci et al. (2023) propose a two-echelon delivery system utilizing multiple trucks and two types of drones-small and large-for delivering relief items in uncertain environments. The trucks operate as mobile depots, launching drones to reach collection points where demand and road conditions are uncertain. Their extensive computational analysis reveals that increasing the number of launch points and small drones significantly reduces unsatisfied demand, whereas expanding the number of depots or large drones has a comparatively smaller impact. Similarly, Liu et al. (2021) develop a sustainable urban distribution network integrating vans and driverless delivery robots. Vans, acting as FE vehicles, transport packages from central depots to satellite facilities (local depots) and return after servicing all assigned satellites. From these satellites, driverless delivery robots complete last-mile deliveries, returning to the same satellite upon completion. Their findings highlight the effectiveness of this system in minimizing costs related to penalties, transshipments, and emissions, thereby achieving sustainable last-mile delivery.

2.2 FE Vehicle and SE Vehicle Co-delivery

In last-mile delivery systems, due to range limitations related to the finite battery capacity or communication range of the SE vehicle

and the desire to limit the number of warehouses due to economies of scale, FE vehicles are deployed alongside SE vehicles to deliver goods to customers to improve the quality of service, and reduce transportation cost or delivery time. Considering the delivery mode and the number of delivery vehicles, we classify the subproblem into two categories: 1) Two-Echelon Traveling Salesman Problem (2ETSP), and 2) Two-Echelon Vehicle Routing Problem (2EVRP). Detailed overviews of these categories are provided in Tables 3, 4 and 5. In the following subsections, we discuss each attribute in detail.

2.2.1 2ETSP and its Variants

An important type of 2ETSP is the Flying Sidekick Traveling Salesman Problem (FSTSP), first introduced by Murray and Chu (2015). The FSTSP extends the traditional TSP by incorporating a two-echelon vehicle delivery system, where a single FE vehicle is equipped with an SE vehicle (e.g., a drone). The SE vehicle can be deployed en route to deliver parcels but must return to the FE vehicle due to its limited flight endurance. Building on this concept, Jeong et al. (2019) study a truck-drone hybrid delivery system to minimize customer service time, particularly focusing on the impact of parcel weight on drone energy consumption and restricted flying areas. They propose a two-phase constructive and search heuristic algorithm to provide computationally efficient solution of real-world problems. Dell'Amico et al. (2019) address the FSTSP as an operational optimization problem where a truck, accompanied by a drone, collaborates to deliver parcels to customers in the minimum completion time. They develop a Branch-

and-Cut (*B&C*) method with a set of effective inequalities, which enhances solution accuracy and computational efficiency. Extending this work, [Dell'Amico et al. \(2021\)](#) introduce a branch-and-bound algorithm tailored for small instances (up to 15 customers) and a heuristic approach that leverages the branch-and-bound method as a subroutine to address larger instances. Considering drone payload capacity and battery power constraints, [Boccia et al. \(2021\)](#) propose a compact integer linear programming formulation for the FSTSP, which overcomes the drawbacks of the big-M constraints required to model the synchronization issue. Their *B&C* algorithm, tested on the same benchmark instances as [Dell'Amico et al. \(2019\)](#), demonstrates comparable or superior computational performance, emphasizing improvements in solving synchronization-related issues. In a multi-drop FSTSP context, [Mara et al. \(2022\)](#) introduce a novel mathematical model that accounts for setup times associated with drone launches and retrievals. To optimize total completion time, they propose a heuristic based on Adaptive Large Neighborhood Search (ALNS), showcasing its effectiveness in handling operational complexities unique to multi-drop scenarios.

Another type of 2ETSP closely related to the FSTSP introduces two flexibility elements: each node can be visited more than once by the FE, and the SE may be launched and return to the same position. The second element is highly significant since it allows the effective management of deliveries to customers located far from others, as found in low housing density and rural areas, and the presence of no-fly zone areas. Based on the number of FE and

SE vehicles involved, three primary delivery modes emerge, which are reviewed in the following.

Single FE and Single SE. The integration of a single FE and SE introduces significant cost-saving opportunities compared to truck-only operations. [Agatz et al. \(2018\)](#) highlight this potential by analyzing a scenario where the drone, being faster than the truck, rejoins it at its release node. Their study provides both theoretical insights and empirical evidence that demonstrate substantial cost reductions. Building on this foundation, [Ha et al. \(2018\)](#) focus on optimizing operational costs in the Truck-and-Drone Scheduling Problem (TSPD). They formulate the problem as a Mixed-Integer Linear Programming (MILP) model and address its computational complexity using a Greedy Randomized Adaptive Search Procedure (GRASP). In contrast to cost minimization, [Roberti and Ruthmair \(2021\)](#) explore completion time minimization in the TSPD, proposing a compact MILP model and solving it via an exact Branch-and-Price (*B&P*) approach. This shift in objective exemplifies the flexibility of TSPD formulations to address diverse operational priorities. Further advancing the problem, [El-Adle et al. \(2023\)](#) focus on synchronizing truck-drone routing to minimize return times to the depot. They propose a Variable Neighborhood Search (VNS) heuristic, which effectively addresses the challenge of coordinating drone cycles with truck movements. [Gonzalez-R et al. \(2020\)](#) extend the TSPD by allowing drones to visit multiple customers between consecutive rendezvous with the truck. Their heuristic demonstrates strong performance in minimiz-

Table 3 Summary of the Main Features of the 2ETSP

Authors	Number of		Objective	SE	SE	SE	Time windows	Delivery class
	FE	SE		energy	capacity	multiple visits		
Agatz et al. (2018)	1	1	min-cost	✓	✓			(d)
Ha et al. (2018)	1	1	min-cost	✓				(d)
Jeong et al. (2019)	1	1	min-time	✓				(d)
Dell'Amico et al. (2019)	1	1	min-time	✓				(d)
Kitjacharoenchai et al. (2019)	n	m	min-time					(e)
Gonzalez-R et al. (2020)	1	1	min-time	✓		✓		(d)
Wang et al. (2020)	1	1	min-time & cost	✓	✓			(d)
Murray and Raj (2020)	1	m	min-time	✓	✓	✓		(d)
Raj and Murray (2020)	1	m	min-time	✓	✓	✓		(d)
Boccia et al. (2021)	1	1	min-time	✓	✓			(d)
Cavani et al. (2021)	1	m	min-time		✓			(d)
Dell'Amico et al. (2021)	1	1	min-time					(d)
Roberti and Ruthmair (2021)	1	1	min-time	✓				(d)
Mara et al. (2022)	1	1	min-time	✓		✓		(d)
Salama and Srinivas (2022)	1	m	min-time		✓			(d)
Bai et al. (2023)	1	1	min-time	✓	✓	✓		(d)
El-Adle et al. (2023)	1	m	min-time					(d)
Tinic et al. (2023)	1	∞	min-cost	✓				(d)
Yang et al. (2023)	1	1	max-profit	✓	✓		✓	(d)
Kloster et al. (2023)	n	m	min-time	✓				(e)

ing the makespan, particularly for large-scale delivery systems, emphasizing scalability. In scenarios where drone payload and operating range are critical, Bai et al. (2023) investigate task assignment strategies for trucks and high-capacity drones. They propose algorithms to plan routes either sequentially or simultaneously, optimizing delivery time while accounting for drone constraints. Meanwhile, Yang et al. (2023) introduce a novel perspective by focusing on profit maximization rather than cost or time. Their work incorporates customer-defined time windows and leverages a B&P

approach to generate optimal solutions, addressing market-driven logistics requirements.

Single FE and Multiple SEs. The single FE and multiple SEs configuration is explored by Murray and Raj (2020), who propose a last-mile delivery system involving a single truck and multiple heterogeneous UAVs. Their MILP formulation incorporates detailed queue scheduling, achieving high-quality solutions for realistic problems. Raj and Murray (2020) further extend this by treating UAV speeds as decision variables, proposing a three-phase heuristic to adjust drone speeds dynamically, mini-

mizing makespan in the process. Cavani et al. (2021) present a compact MILP model for delivery systems with a single traditional vehicle and multiple drones working in tandem to minimize the time required to complete all the deliveries and return to the depot, supplemented with various families of valid inequalities to enhance solution quality. Luo et al. (2021) discuss a multi-visit TSP with multiple drones, which not only considers multiple deliveries but also adopts an endurance model based on the drone's own weight, its payload, as well as the flight time. They propose a two-echelon route generating algorithm that efficiently derives optimal, near-optimal, or reasonable vehicle schedules when the MILP fails to produce optimal schedules within a reasonable time frame. Salama and Srinivas (2022) address the coordination of a truck and multiple heterogeneous drones for last-mile package delivery and does not restrict drone launch and recovery operations to be at customer locations. A new MILP model is formulated to solve this problem optimally with the objective of minimizing delivery completion time. Tinic et al. (2023) introduce the minimum cost TSP with multiple drones, where a truck and multiple drones work in synchronization to deliver parcels to customers. They provide flow based and two cut based MILP formulations strengthened with valid inequalities.

Multiple FEs and SEs. Expanding upon the framework of the FSTSP to accommodate multiple trucks and drones, Kitjacharoenchai et al. (2019) introduce an Mixed Integer Programming (MIP) formulation aimed at minimizing the arrival times of trucks and drones at the depot after delivery. They also pro-

pose an insertion heuristic-based algorithm to manage large-scale problem instances effectively. Kloster et al. (2023) address the multiple TSP with multiple drone stations, considering the load-dependent energy consumption of drones. Their MILP model accounts for drones that can launch from depots, trucks, or drone stations and return to any of these locations. They also develop three algorithms to solve this problem, optimizing both delivery time and energy usage.

2.2.2 2EVRP and its Variants

The 2EVRP is a variant of the 2ETSP that involves the collaboration of multiple trucks and drones to provide services to a given set of nodes. There are several 2EVRP variants that incorporate constraints in real-life vehicle routing applications, such as the Capacitated Vehicle Routing Problem (2ECVRP), Vehicle Routing Problem with Time Windows (2EVRPTW), Multi-Depot Vehicle Routing Problem (2EMDVRP), Multi-visit Vehicle Routing Problem (2EMVVRP) and Vehicle Routing Problem for Humanitarian Logistics (2EVRPHL).

2ECVRP. The capacity constraint includes both the capacity constraints of FE vehicles and SE vehicles. For SE vehicles, Karak and Abdelghany (2019) study the use of multiple drones on a single vehicle, with each drone having a limited flying range and loading capacity. They solve instances with up to 100 customers and 24 stations. Simoni et al. (2020) focus on robots with maximum parcel capacities, proposing an MILP for small instances and a heuristic for larger instances. Moshref-Javadi et al. (2020a) and Moshref-Javadi et al. (2020b) assume that UAVs have limited capacity, so that each UAV can serve only one cus-

Table 4 Summary of the Main Features of the 2EVRP

Authors	Number of		Objective	SE	SE	SE	Time windows	Delivery class
	FE	SE		energy	capacity	multiple visits		
Chiang et al. (2019)	n	m	min-cost		✓	✓		(e)
Karak and Abdelghany (2019)	1	m	min-cost	✓	✓			(d)
Sacramento et al. (2019)	n	m	min-cost		✓		✓	(e)
Schermer et al. (2019a)	n	m	min-time					(e)
Schermer et al. (2019b)	n	m	min-time					(e)
Wang and Sheu (2019)	n	m	min-cost	✓	✓	✓		(e)
Kitjacharoenchai et al. (2020)	n	m	min-time	✓	✓	✓		(e)
Moshref-Javadi et al. (2020a)	1	m	min-time	✓	✓			(d)
Moshref-Javadi et al. (2020b)	1	m	min-time	✓	✓	✓		(d)
Simoni et al. (2020)	1	1	min-time		✓	✓		(d)
Chen et al. (2021a)	n	m	min-time	✓	✓		✓	(e)
Chen et al. (2021b)	n	m	min-time	✓	✓		✓	(e)
			min-cost;					
Luo et al. (2021)	1	m	max-				✓	(d)
			satisfaction					
Kang and Lee (2021)	1	m	min-time	✓				(d)
Tamke and Buscher (2021)	n	m	min-time	✓	✓			(e)
Zhang et al. (2021)	1	1	max-profit	✓	✓	✓		(d)
Gu et al. (2022)	n	m	min-cost	✓	✓	✓		(e)
Heimfarth et al. (2022)	1	m	min-cost				✓	(d)
Lei et al. (2022)	n	m	min-cost					(e)
Lu et al. (2022)	n	m	min-time & equity	✓	✓	✓		(e)
Masmoudi et al. (2022)	n	m	max-profit	✓	✓	✓	✓	(e)
Wang et al. (2022)	n	m	min-cost		✓		✓	(e)
Wu et al. (2022)	n	m	min-time	✓	✓	✓		(e)

tomers per delivery, resulting in a significant reduction in wait times compared to the truck-only model. Similarly, Wang et al. (2020) suggest that the volume and weight of parcels for certain customers may exceed the capacity of the drone, requiring them to be served exclusively by truck. In Wu et al. (2022), the payload weight of the drone during delivery is limited and the capacity of trucks is sufficient

to support the entire delivery, highlighting the importance of payload capacity in minimizing delivery time. Kang and Lee (2021), Tamke and Buscher (2021) and Thomas et al. (2023) also assume that the truck has sufficient capacity to meet all requirements. Kang and Lee (2021) require each drone to have a specific speed and battery capacity and each drone to carry a single demand at a time. They use a

logic-based Benders decomposition approach to minimize the total of truck trips and waiting times for drones to return after deliveries, outperforming state-of-the-art solvers. [Tamke and Buscher \(2021\)](#) address the constraint that a drone can only deliver to one customer at a time due to weight limitations. They solve instances with up to 30 customers using a customized *B&C* algorithm. [Thomas et al. \(2023\)](#) consider a heterogeneous drone fleet characterized by payload carrying capacity and they find that the min-cost model has lower usage of low payload and heavy payload drones. For FE vehicles, [Lei et al. \(2022\)](#) assume that trucks have limited capacity and impose a constraint that a drone can only visit one customer at a time. They use a dynamic artificial bee colony algorithm to minimize the total operating cost, and find new optimal solutions for 37 instances of the 2EVRP. For FE and SE vehicles' capacity constraints, [Chiang et al. \(2019\)](#) focus on package delivery using FE vehicles equipped with UAVs. They demonstrate cost-effective and environmentally friendly solutions by employing drones alongside vehicles. [Sacramento et al. \(2019\)](#) propose an ALNS metaheuristic to minimize truck operating costs, considering truck capacity constraints and the maximum load capacity of a drone. Their sensitivity analysis shows that adjusting the maximum payload capacity of the drone allows for a greater reduction in fuel costs. [Wang and Sheu \(2019\)](#), [Wang et al. \(2022\)](#), [Najy et al. \(2023\)](#) and [Xia et al. \(2023\)](#) highlight the importance of payload capacity in delivery modes by considering capacity constraints for both trucks and drones, thus emphasizing the effect of capacity on delivery costs.

2EVRPTW. The consideration of time window constraints is critical in the 2EVRP. Several studies have extended the 2EVRP by incorporating hard/soft time windows, where hard time windows define absolute earliest and latest service times, and soft time windows denote preferred service times within the hard constraints. [Chen et al. \(2021ab\)](#) focus specifically on integrating delivery robots into the VRP with hard time windows. In their study, [Chen et al. \(2021a\)](#) assign predefined hard time windows to each customer and prohibit deliveries outside of these windows. Their MILP approach proves effective for up to 50 customers, highlighting the benefits of using delivery robots for last-mile delivery, especially for customers with hard time windows. Meanwhile, [Chen et al. \(2021b\)](#) consider a scenario where all customers and a single depot have time windows, and delivery can only occur within these windows, even if the delivery resource arrives before the lower bound. Their results emphasize the operational challenges of route planning and demonstrate the potential of self-driving delivery robots in urban areas. [Masmoudi et al. \(2022\)](#) introduce the VRPD with multi-package payload compartments, with the goal of maximizing total profit while respecting hard customer time windows. Their adaptive multi-start simulated annealing (SA) algorithm efficiently handles instances with up to 200 customers, despite the challenges posed by time windows. [Yu et al. \(2022\)](#) consider pickup and delivery operations along with hard customer time windows in a two-echelon van-based robotic system. They formulate an MIP and propose an ALNS algorithm to minimize the total travel

Table 5 Summary of the Main Features of the 2EVRP(Continued)

Authors	Number of		Objective	SE	SE	SE	Time	Delivery
	FE	SE		energy	capacity	multiple visits		
Yu et al. (2022)	n	m	min-cost	✓	✓	✓	✓	(e)
Kuo et al. (2023)	n	m	min-time	✓			✓	(e)
Lu et al. (2023)	n	m	min-time & equity	✓	✓	✓	✓	(e)
Meng et al. (2023)	n	m	min-cost	✓	✓	✓		(e)
Najy et al. (2023)	1	1	min-cost		✓			(d)
Ostermeier et al. (2023)	1	m	min-cost				✓	(d)
Rave et al. (2023)	n	m	min-cost					(e)
Thomas et al. (2023)	1	m	min-time	✓	✓			(d)
Xia et al. (2023)	n	m	min-cost	✓	✓	✓		(e)
Yin et al. (2023a)	n	m	min-cost	✓	✓	✓	✓	(e)
Yin et al. (2023b)	n	m	min-cost	✓	✓	✓		(e)

cost, with time window constraints playing a critical role due to their significant impact on the solution results. Heimfarth et al. (2022) consider scenarios where each customer has a delivery window defined by a deadline and a window length. Their general VNS algorithm is competitive, outperforming existing truck-and-robot routing algorithms at runtime while maintaining solution quality comparable to a robot-only delivery approach. Yin et al. (2023a) develop an extended branch-and-price-and-cut algorithm that incorporates a bounded bidirectional labeling algorithm to solve the 2EVRP with hard time windows, where each customer is associated with a time window that specifies the time interval within which they are available for delivery service in the planning horizon. Kuo et al. (2023) introduce the 2EVRPTW model, where the service level is calculated using a soft time window system. Their goal is to minimize the makespan and carbon emissions of the de-

livery route. They use NSGA-II to solve the problem and test the proposed algorithm with problems of different scales.

2EMDVRP. Unlike scenarios where SE vehicles can be launched and retrieved only at vertices, Schermer et al. (2019a) study a problem where drones can also be deployed at discrete points along each arc. They formulate this problem as an MILP and introduce valid inequalities to improve the performance of the solver. Furthermore, they extend their study to allow cyclic drone operations in the VRPD, formulate it as an MILP solvable by standard methods, and introduce sets of valid inequalities to improve solver efficiency (Schermer et al. 2019b). Rave et al. (2023) study the two-echelon location routing problem with drones, aiming to find the cost-optimal fleet mix of trucks and drones as well as the number and locations of microdepots. They develop a problem-specific ALNS that generates near-optimal solutions. Their computational study shows that

while it is mostly profitable to launch drones from microdepots instead from trucks, drones launched from trucks are increasingly preferred as the drone range increases or customer eligibility for drone deliveries decreases.

2EMVVRP. Several studies extend the 2EVRP with multi-visit capabilities, allowing SE vehicles to serve multiple customers per flight. [Kitjacharoenchai et al. \(2020\)](#) explore a synchronized truck-drone 2EVRP, where multiple drones can fly from a truck, serve one or more customers, and return to the same truck to replace batteries and retrieve packages. They develop a two-stage algorithm for the multi-visit 2EVRP that handles instances with 70 customers. The results of the experiment demonstrate the potential benefit of implementing trucks and drones together with the multiple drop feature to reduce the total travel time significantly. [Gu et al. \(2022\)](#) introduce a hybrid iterative local search (ILS) heuristic and variable neighborhood descent (VND) procedure to address the 2EVRP with multiple visits, incorporating a battery endurance model for drones based on payload and time constraints. Their algorithm obtains promising computational results in reasonable computation time. [Meng et al. \(2023\)](#) consider a variant of the combined truck-drone routing problem, where drones serve multiple customers and provide both pickup and delivery services in each flight. Their algorithm outperforms two benchmark heuristics in numerical experiments, demonstrating the significant advantage of allowing multiple visits and both pickup and delivery operations in each drone flight. [Ostermeier et al. \(2023\)](#) extend the truck-and-robot delivery system by incorporat-

ing robot depots to facilitate the simultaneous routing of multiple trucks. The robots are able to visit customers within the same neighborhood at different times, with defined delivery windows. Through experiments, they demonstrate a 62% savings compared to traditional truck delivery methods.

2EVRPHL. Humanitarian logistics involves the efficient management of the flow of supplies, materials, and information to help disaster victims and alleviate suffering. Its key features include the unpredictability of demand, cost, and travel time, due to lack of reliable data, which poses significant challenges for post-disaster relief operations. [Zhang et al. \(2021\)](#) develop a collaborative truck-and-drone system for post-disaster assessment in humanitarian relief. They formulate the problem as an MILP model and decompose it into a path-based master problem and two subproblems to apply column generation. [Lu et al. \(2022\)](#) propose an MILP model for the multi-objective humanitarian pickup and delivery VRP with drones, which aims to minimize the routing time and maximize the fulfillment rate. They propose a hybrid multi-objective evolutionary algorithm and ant colony algorithm to solve this problem effectively. In addition, they study the impact of time-varying weather conditions on synchronous delivery in humanitarian logistics, addressing a multi-objective optimization problem using a hybrid evolutionary algorithm ([Lu et al. 2023](#)). [Yin et al. \(2023b\)](#) develop an extended branch-and-price-and-cut algorithm to find the optimal synthetic route that minimizes the sum of dispatching costs, transportation costs, and worst-case delay penalty costs. Their goal is to ensure

Table 6 Alternative Delivery Modes

$\sum_{s \in C_E} \sum_{(i,j) \in A} \sum_{k_2 \in K_2} y_{ijs}^{k_2} = 1$	SE vehicle delivery only
$\sum_{(i,j) \in A} \sum_{k_1 \in K_1} x_{ij}^{k_1} + \sum_{s \in C_E} \sum_{(i,j) \in A} \sum_{k_2 \in K_2} y_{ijs}^{k_2} = 1$	FE vehicle and SE vehicle co-delivery

that the demand of each affected area is satisfied and that each affected area is served exactly once.

2.3 Summary

In a two-echelon delivery system, goods can be transported to customers either through SE vehicle-only delivery or via coordinated co-delivery by both FE and SE vehicles. Table 6 outlines the constraints that govern the operational scenarios for these vehicles, ensuring that the requirements of each model are met while maintaining feasibility and efficiency. The sets K_1 and K_2 represent FE and SE vehicles, respectively, and the formulation incorporates two types of decision variables tailored to the distinct delivery models. For the SE vehicle-only delivery model, a binary routing variable $y_{ijs}^{k_2}$ is defined to indicate whether SE vehicle $k_2 \in K_2$ traverses the arc $(i, j) \in A$, where $A = \{(i, j) | i, j \in N, i \neq j\}$, and N denotes the set of all nodes in the system. Furthermore, the SE vehicle must return to the FE vehicle at node $s \in C_E$, where C_E denotes the set of customers and the drop-off/local depot as the terminal location. For the co-delivery model involving both FE and SE vehicles, a binary routing variable $x_{ij}^{k_1}$ is introduced to denote whether the FE vehicle $k_1 \in K_1$ traverses arc $(i, j) \in A$, where $A = \{(i, j) | i, j \in N, i \neq j\}$ and N again represents the set of all nodes. This model ensures coordination between the FE and SE vehicles to optimize delivery operations.

Table 7 classifies studies that utilize the co-

operated concept and differ mainly on the policy followed by the FE vehicle after dispatching the SE vehicle. In one policy, the FE vehicle waits at the same local depots/drop-off points for the SE vehicle to return, which we call *dispatch-wait-collect*. This approach prioritizes coordination between the FE and SE vehicles, ensuring that both vehicles are synchronized in their movements and can proceed together after the SE vehicle completes its task. This policy may be beneficial in scenarios where close collaboration and real-time coordination between the vehicles is crucial, such as in time-sensitive deliveries, or where precise synchronization is required for seamless operations. Alternatively, the FE vehicle continues its tour but later collects the deployed SE vehicles, which we call *dispatch-move-collect*. This policy allows the FE vehicle to optimize its route and maximize its coverage area while still ensuring that the SE vehicles are eventually collected and returned to the FE vehicle. This policy may be advantageous in scenarios where the FE vehicle's primary objective is to cover as much ground as possible efficiently while maintaining flexibility in its itinerary. As another alternative, *Dispatch-leave* means that the FE vehicle continues its tour without needing to collect the SE vehicles. This policy is characterized by the FE vehicle's independence from the SE vehicle's movements, allowing the FE vehicle to proceed with its operations without waiting for or retrieving the SE vehicle. This policy is suitable for scenarios where the

FE vehicle's tasks are independent of or unaffected by the actions of the SE vehicle, enabling greater autonomy and flexibility in the FE vehicle's routing decisions.

3. Methodologies

In this section, Tables 8, 9 and 10 provide details about mathematical models, algorithm types, and instances used for computational tests. This categorization aids in understanding the diverse approaches taken by different methodologies. Additionally, we explore the application of two-echelon vehicle operations in both simulation and practical settings, offering insights into the practical implications and real-world effectiveness of the proposed solution methodologies.

3.1 Two-echelon Vehicle Operations in Simulation

Several solution methodologies have been proposed for the 2ECOP, and these can be broadly categorized into two groups: exact and heuristic methods, as shown in Tables 8, 9 and 10. The NP-hard nature of the 2ECOP makes it impractical to employ off-the-shelf MILP solvers, for example Gurobi or CPLEX for larger instances of the problem. As a result, heuristics have gained popularity for solving the 2ECOP due to their efficient running time.

We identify the best-performing exact algorithms and heuristics by listing their reported computational results on customer instances. In Tables 8, 9 and 10, these methods demonstrate effectiveness at finding optimal solutions on instances with up to 50 customers and prove to be efficient heuristics for instances with up to 250 customers.

Within the existing literature, a prevalent approach involves decomposing the 2ECOP into two subproblems: the location-allocation problem and the vehicle routing problem. These subproblems are sequentially and iteratively solved using a 'k-prototype cluster' and an improving algorithm. A recurring pattern in most two-echelon delivery heuristics is to generate an initial solution by dividing the two-echelon delivery into separate subproblems, one for each echelon. Subsequently, the VRP/TSP related to the second echelon is solved first. Once the routing problem for the second echelon is solved, all demands become known, allowing the construction of the first echelon routes. This process can be performed using various methods, including single-solution-based approaches such as local neighborhood search, variable neighborhood descent, variable neighborhood search, tabu search, large neighborhood search, or adaptive large neighborhood search. Alternatively, population-based methods such as genetic algorithms can be used.

3.2 Two-echelon Vehicle Operations in Practice

The inception of routing in a two-echelon distribution system dates back to Jacobsen and Madsen (1980), who address a combined location/routing problem in the distribution of newspapers in western Denmark. They propose three heuristics to minimize costs.

Several studies employ their proposed algorithm in practical urban distribution scenarios. Moshref-Javadi et al. (2020a) implement a truck and drone routing algorithm in a practical e-commerce delivery scenario situ-

Table 7 Cooperation Mode of FE and SE Vehicles

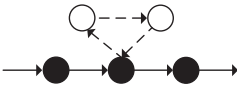
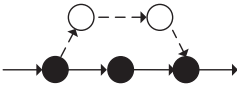
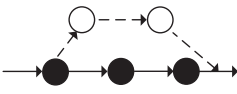
Modes	Figures	Authors
Dispatch-wait-collect		Boysen et al. (2018); Chang and Lee (2018); Moshref-Javadi et al. (2020a); Chen et al. (2021a); Chen et al. (2021b); Liu et al. (2021); Kang and Lee (2021); Zhang et al. (2021); Heimfarth et al. (2022); Rave et al. (2023); Bai et al. (2023); Dukkanci et al. (2023); El-Adle et al. (2023); Kloster et al. (2023); Momeni et al. (2023); Ostermeier et al. (2023); Tinic et al. (2023); Thomas et al. (2023); Yang et al. (2023)
Dispatch-move-collect		Agatz et al. (2018); Ha et al. (2018); Chiang et al. (2019); Dell'Amico et al. (2019); Jeong et al. (2019); Karak and Abdelghany (2019); Kitjacharoenchai et al. (2019); Sacramento et al. (2019); Schermer et al. (2019ab); Wang and Sheu (2019); Chung et al. (2020); Gonzalez-R et al. (2020); Kitjacharoenchai et al. (2020); Moshref-Javadi et al. (2020b); Murray and Raj (2020); Poikonen and Golden (2020); Raj and Murray (2020); Simoni et al. (2020); Boccia et al. (2021); Cavani et al. (2021); Dell'Amico et al. (2021); Luo et al. (2021); Roberti and Ruthmair (2021); Tamke and Buscher (2021); Zhang et al. (2021); Gu et al. (2022); Lei et al. (2022); Lu et al. (2022); Mara et al. (2022); Masmoudi et al. (2022); Salama and Srinivas (2022); Wang et al. (2020); Wang et al. (2022); Wu et al. (2022); Yu et al. (2022); Bai et al. (2023); El-Adle et al. (2023); Kuo et al. (2023); Lu et al. (2023); Meng et al. (2023); Najy et al. (2023); Tinic et al. (2023); Thomas et al. (2023); Xia et al. (2023); Yang et al. (2023); Yin et al. (2023a); Yin et al. (2023b)
Dispatch-leave		Boysen et al. (2018); Alfandari et al. (2022); Heimfarth et al. (2022); Meng et al. (2023); Ostermeier et al. (2023); Thomas et al. (2023); Wang et al. (2020)

Table 8 Mathematical Models, Exact Approach and Problem Instances

Authors	Mathematical Model	Approach	Largest Instance	Case Study
Alfandari et al. (2022)	✓	Benders decomposition	50 customers	No
Dukkanci et al. (2023)	✓	Scenario decomposition	25 districts	Yes
Dell'Amico et al. (2019)	✓	Branch & cut	13 customers	No
Wang and Sheu (2019)	✓	Branch & price	15 customers	No
Boccia et al. (2021)	✓	Column and row generation	20 customers	No
Cavani et al. (2021)	✓	Branch & cut	24 customers	No
Kang and Lee (2021)	✓	Benders decomposition	40 customers	No
Roberti and Ruthmair (2021)	✓	Branch & price	39 customers	No
Tamke and Buscher (2021)	✓	Branch & cut	30 customers	No
Najy et al. (2023)	✓	Branch & cut	50 customers	No
Tinic et al. (2023)	✓	Branch & cut	30 customers	No
Xia et al. (2023)	✓	Branch and price and cut	10 customers	No
Yang et al. (2023)	✓	Branch & price	20 customers	Yes
Yin et al. (2023a)	✓	Branch and price and cut	45 customers	No
Yin et al. (2023b)	✓	Branch and price and cut	45 customers	No
Dell'Amico et al. (2021)	✓	Branch & bound / Iterative heuristic ε -constraint /	15 customers for exact; 229 customers for heuristic	No
Liu et al. (2021)	✓	hybrid artificial immune algorithm	100 customers	Yes

ated in São Paulo, Brazil, effectively minimizing customer waiting times. Through their comprehensive case study analyses, their algorithm demonstrates proficient resolution of real-world problem instances. Additionally, they show that the number of UAVs significantly influences the waiting time performance of delivery systems. Mara et al. (2022) model the logistics network of a retail company in the urban setting of Yogyakarta, Indonesia. As the scale of the problem increases, the ALNS consistently yields solutions of higher quality. Wang et al. (2022) randomly generate the locations of depot and customer nodes across the

map of Chongqing City, China. They evaluate the impact of road traffic on optimization outcomes, utilizing real-time road traffic data through an iterative local search heuristic algorithm. Their findings indicate that the truck-drone solution enhances delivery punctuality, substantially diminishes trucking mileage, and effectively mitigates terrain constraints. Yang et al. (2023) construct a dataset comprising 20 customers sourced from Amazon delivery data within the greater Boston area, with each location denoted by a latitude/longitude pair. This case study illustrates that the problem can be efficiently solved to provable optimal-

Table 9 Mathematical Models, Heuristic Approach and Problem Instances

Authors	Mathematical Model	Approach	Largest Instance	Case Study
Zhang et al. (2021)	✓	Column generation based heuristic	25 nodes	yes
Boysen et al. (2018)	✓	Local search	40 customers	no
Poikonen and Golden (2020)	✓	Three-phase algorithm	50 customers	no
Momeni et al. (2023)	✓	Two-phase algorithm	100 customers	yes
Agatz et al. (2018)	✓	First-cluster second heuristic	100 customers	no
Ha et al. (2018)	✓	Greedy randomized adaptive search algorithm	100 customers	no
Chiang et al. (2019)	✓	Genetic algorithm	500 customers	no
Jeong et al. (2019)	✓	Evolutionary-based heuristic	50 customers	no
Karak and Abdelghany (2019)	✓	Hybrid Clarke and Wright heuristic	100 customers	no
Kitjacharoenchai et al. (2019)	✓	Adaptive insertion algorithm	50 customers	no
Sacramento et al. (2019)	✓	ALNS metaheuristic	100 customers	no
Schermer et al. (2019a)	✓	Hybrid VNS/Tabu search algorithm	50 customers	no
Schermer et al. (2019b)	✓	Matheuristic	100 customers	no
Gonzalez-R et al. (2020)	✓	Iterated greedy heuristic	250 customers	no
Kitjacharoenchai et al. (2020)	✓	ALNS	80 customers	no
Murray and Raj (2020)	✓	Three-phased heuristic solution approach	100 customers	no
Wang et al. (2020)	✓	Improved NSGA-II	100 customers	no
Moshref-Javadi et al. (2020a)	✓	Metaheuristic	101 customers	yes
Moshref-Javadi et al. (2020b)	✓	Hybrid Tabu/SA search algorithm	100 customers	yes
Raj and Murray (2020)	✓	Three-phased algorithm	100 customers	no
Simoni et al. (2020)	✓	Local search	500 customers	no

Table 10 Mathematical Models, Heuristic Approach and Problem Instances (Continued)

Authors	Mathematical Model	Approach	Largest Instance	Case Study
Chen et al. (2021a)	✓	Metaheuristic	200 customers	no
Chen et al. (2021b)	✓	Metaheuristic	50 customers	no
Luo et al. (2021)	✓	Hybrid multi-objective optimization approach	80 customers	no
Gu et al. (2022)	✓	Hybrid ILS/VND	200 customers	no
Heimfarth et al. (2022)	✓	General VNS	100 customers	no
Lei et al. (2022)		Dynamical artificial bee colony	200 customers	no
Lu et al. (2022)	✓	Hybrid multi-objective evolutionary algorithm	100 customers	yes
Mara et al. (2022)	✓	ALNS	200 customers	yes
Masmoudi et al. (2022)	✓	Adaptive multi-start simulated annealing algorithm	100 customers	no
Salama and Srinivas (2022)	✓	Hybrid SA/VNS search algorithm	50 customers	no
Wang et al. (2022)	✓	Iterative local search heuristic algorithm	200 customers	yes
Wu et al. (2022)	✓	VND	200 customers	yes
Yu et al. (2022)	✓	ALNS	50 customers	yes
Bai et al. (2023)	✓	Decoupled task assignment algorithm	150 customers	no
El-Adle et al. (2023)	✓	VNS	100 customers	no
Kloster et al. (2023)	✓	Decomposition-based matheuristic	25 instances	no
Kuo et al. (2023)	✓	NSGA-II	150 customers	no
Lu et al. (2023)	✓	Hybrid multi-objective evolutionary algorithm	200 customers	yes
Meng et al. (2023)	✓	Improved simulated annealing	80 customers	no
Ostermeier et al. (2023)	✓	Truck-and-Robot clustering and routing heuristic	50 customers	no
Rave et al. (2023)	✓	ALNS	200 customers	no
Thomas et al. (2023)	✓	Optimization-based two-phase heuristic	100 customers	no

ity within a reasonable timeframe utilizing the proposed *B&P* solution method. Considering the economic feasibility and battery constraints of driverless delivery robots, Liu et al. (2021) select Miami County, one of the most densely populated areas in Florida, as a benchmark for node location generation in parcel distribution. Their results demonstrate that the hybrid artificial immune algorithm outperforms NSGA-II for practical applications.

Some scholars employ their proposed algorithms in practical distribution scenarios following seismic events. Dukkanci et al. (2023) undertake a case study centered on a post-disaster scenario in Istanbul, Turkey, motivated by the anticipation of an earthquake in the region. Their findings demonstrate the effectiveness of their scenario decomposition algorithm as a robust solution approach, particularly when applied to a small dataset, in contrast to the deterministic equivalent problem formulation. Yin et al. (2023b) employ a robust MIP model to address challenges stemming from the Wenchuan earthquake in China. Within this context, they identify four nodes connected by severely damaged roads, designating them as accessible exclusively by drones. Their findings underscore the advantages of a collaborative truck-drone transport mode compared to a truck-only approach.

Taking the COVID-19 epidemic in Wuhan as an example, Lu et al. (2022) compare the truck-drone cooperated delivery model with the truck-only delivery and drone-only delivery models, and find that their model has advantages in the delivery efficiency of anti-epidemic materials. Later, Lu et al. (2023) take the distribution of emergency supplies distri-

bution in the flood-stricken area of Gongyi City, China in July 2021 as a sample case to verify the applicability of the proposed model in practice. They find that weather conditions play an important role in the truck-drone synchronized delivery decision by comparing the results of the basic model and extended model.

In summary, these case studies demonstrate the diverse applications and effectiveness of routing algorithms in two-echelon distribution systems across different scenarios, industries, and geographical locations. These results bring two-echelon delivery systems closer to practice and motivate future research on designing more effective algorithms to support the decision-making process involved therein.

4. Future Research Areas

In this section, we identify possible future research directions based on the literature survey provided in the above sections.

4.1 Extending the Scope of Current Research

The two-echelon vehicle cooperation methods discussed in this work have great potential to improve delivery efficiency and thus enhance the productivity and quality of service. Many research opportunities are identified from the above literature and presented in Table 11. It is important to study the challenges associated with two-echelon vehicle cooperated operations, particularly those involving time windows, multiple vehicles, and multiple depots, within the realms of management and operations research. Furthermore, it is important to enhance the realism of the existing models by integrating dynamic demand considerations,

such as real traffic network characteristics, and incorporating functionalities like robots' battery recharging capabilities. Due to the complexity of optimization of the two-echelon routes, more efficient algorithms need to be developed in future work. Additionally, with technical problems and regulations for environmentally friendly vehicles such as electric vehicles delivery improved, there have been many research studies considering the pollution routing problem. Researchers should develop corresponding models of routing problems considering the features of environmental protection vehicles. Moreover, future studies should expand upon existing models by introducing multiple objectives, including minimizing operational and maintenance costs (encompassing insurance, labor, and fuel expenses), reducing time spans, maximizing profitability, and mitigating carbon emissions.

4.2 New Research Directions for 2ECOP

Based on our findings and insights from the above literature review, we discuss several research opportunities that are of the greatest practical and theoretical interest. In Fig. 3, we summarize future research areas including the environment, technology, human effects, and government.

First, the development of novel multi-objective optimization approaches for the 2ECOP which take environmental factors into account, would be valuable. The environmental impact assessment focuses primarily on exploring strategies to minimize the carbon footprint associated with these systems. This entails considerations such as enhancing energy efficiency and adopting alternative fuel sources

to mitigate environmental impact. Additionally, addressing uncertainty introduces operational flexibility, enabling route modifications during daily operations due to contingencies like weather, demands of customers, or other capacity-related factors. Future research questions can be addressed using multiple objectives, for example the makespan and carbon emissions of the delivery route (see Kuo et al. (2023)), and cost and fuel consumption (see Chiang et al. (2019)). Much remains to be discovered about these complex and important tradeoffs.

Second, focusing on human-centric aspects of two-echelon vehicle systems is identified as a priority research area for the future. In the modern transportation landscape, advances in science and technology continually reshape delivery methods, modes, and scales. Concurrently, there is a shift in delivery modes from centralized distribution to more personalized approaches. As discussed in Section 2, the existing literature often assumes that customer locations serve as both launch and landing nodes for the second vehicle. Future investigations should study network configurations that incorporate customer nodes alongside distinct non-residential rendezvous locations designated specifically for the launch and retrieval of delivery drones. This expansion would enhance the scope and adaptability of two-echelon vehicle cooperated operations.

Third, within the realm of technological advancements, there is a focus on studying the integration of technologies aimed at enhancing the payload capacity, battery power, and energy efficiency of vehicles within the two-echelon system. This involves leverag-

Table 11 Future Topics Proposed by Current Research

Future Topics	Authors
Vehicles styles	Chiang et al. (2019); Liu et al. (2021); Alfandari et al. (2022); Lei et al. (2022); Mara et al. (2022)
Multiple vehicles and multiple drones	Agatz et al. (2018); Ha et al. (2018); Jeong et al. (2019); Sacramento et al. (2019); Kitjacharoenchai et al. (2020); Moshref-Javadi et al. (2020a); Murray and Raj (2020); Liu et al. (2021); Wu et al. (2022)
Technical constraints and considerations	Chang and Lee (2018); Kitjacharoenchai et al. (2019); Murray and Raj (2020); Simoni et al. (2020); Chen et al. (2021ab); Liu et al. (2021); Alfandari et al. (2022); Lei et al. (2022); Srinivas et al. (2022); Bai et al. (2023); Rave et al. (2023); Xia et al. (2023); Yin et al. (2023b)
Dynamic orders and requirements	Jeong et al. (2019); Sacramento et al. (2019); Murray and Raj (2020); Chen et al. (2021ab); Alfandari et al. (2022); Heimfarth et al. (2022); Mara et al. (2022); Wang et al. (2022)
Improved solution approaches	Agatz et al. (2018); Boysen et al. (2018); Ha et al. (2018); Kitjacharoenchai et al. (2019); Sacramento et al. (2019); Schermer et al. (2019ab); Wang and Sheu (2019); Kitjacharoenchai et al. (2020); Murray and Raj (2020); Poikonen and Golden (2020); Simoni et al. (2020); Boccia et al. (2021); Liu et al. (2021); Alfandari et al. (2022); Mara et al. (2022); Wang et al. (2022); Wu et al. (2022); Bai et al. (2023); Kuo et al. (2023); Ostermeier et al. (2023); Yin et al. (2023a)
Diversification of objective functions	Boysen et al. (2018); Jeong et al. (2019); Kitjacharoenchai et al. (2020); Simoni et al. (2020); Mara et al. (2022); Srinivas et al. (2022); Ostermeier et al. (2023)

ing technologies like the Internet of Things (IoT) and artificial intelligence to enhance the capabilities of these vehicles, and thereby to enhance the overall efficiency and flexibility of two-echelon vehicle cooperated operations.

Fourth, it is important to explore the potential benefits and challenges associated with the use of collaborative multi-agent systems in

two-echelon vehicle operations. Specific areas of focus should include the development of algorithms for real-time communication and coordination between drones and ground vehicles to optimize delivery routes, the integration of machine learning techniques to predict and adapt to changing traffic and weather conditions, and the implementation of robust

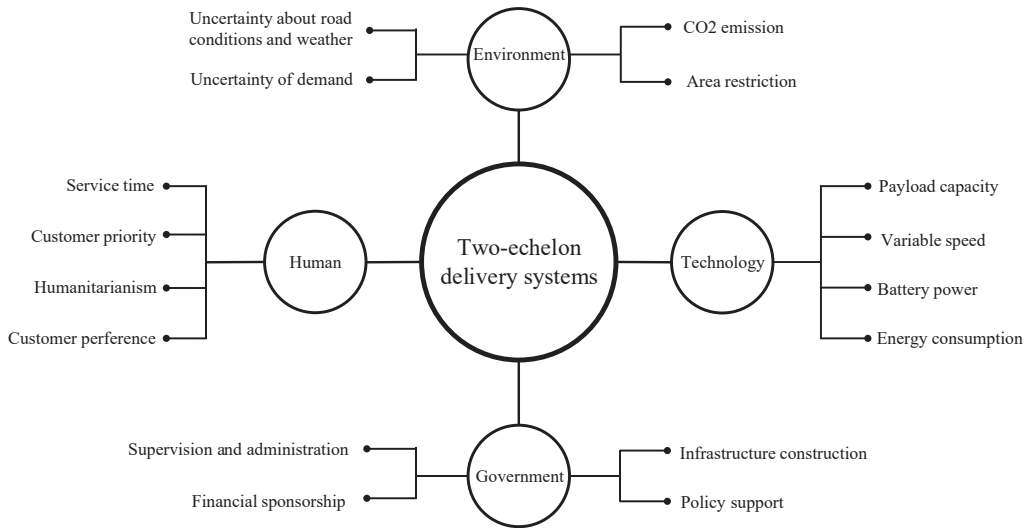


Figure 3 Areas for Future Research

cybersecurity measures to protect the system from potential threats. In addition, exploring economic implications, for example cost-benefit analyses of deploying such systems at scale, and investigating the regulatory and infrastructure requirements needed to support multi-agent collaboration in urban and rural environments are critical to the successful implementation of these advanced logistics systems.

Finally, government financial support, policy initiatives, and infrastructure construction are crucial factors that can facilitate the adoption of two-echelon vehicle distribution systems. In recent years, technological advances have led to a surge in the use of two-echelon vehicles in urban logistics, supply chain management, retail and e-commerce distribution, as well as emergency medical services. Identifying industries and operational contexts where the application of two-echelon vehicles can lead to substantial improvements in supply chain performance, operational cost reduction, and overall logistics efficiency presents valuable opportunities for future study.

5. Concluding Remarks

The integration of different vehicles in both primary and secondary echelons has emerged as a strategic approach to optimize package delivery, resulting in significant cost savings and efficient service delivery. Through a systematic literature review, we survey the existing literature on two-echelon cooperative problems and categorize it into two classes based on whether the FE vehicle delivers packages to customers. Our review identifies several challenging extensions of the 2ECOP, including a heterogeneous fleet of first- and second-echelon vehicles, multiple depots, multiple visits, time windows, and humanitarian considerations. A large number of both exact and heuristic methods have been developed to address the 2ECOP and its variants. To obtain reasonable solutions in a short time, heuristics have been developed. In addition, we propose a unified algorithmic framework for deployment aimed at reducing costs and increasing service efficiency. This comprehensive review and proposed framework contribute to ongoing efforts to address

the complexities of modern package delivery logistics.

Two-echelon delivery systems offer significant benefits to companies operating in specific industries and facing specific logistical challenges. In our review of work on the 2ECOP, we examine real-world-inspired problem variants that aim to bridge the gap between canonical models and practical applications. In practice, two-echelon networks are prevalent in urban logistics, where the goal is to transport goods efficiently in urban areas while considering safety and environmental concerns. Coordinating consolidation opportunities in urban logistics is becoming increasingly important, especially due to the growing number of zero- and low-emission zones in many cities. In addition, two-echelon distribution networks are prevalent in other sectors such as express delivery, grocery and supermarket product distribution, multimodal freight transportation, and e-commerce and home delivery services. Thus, while two-echelon delivery can be highly beneficial for many companies, its suitability and feasibility vary depending on the specific characteristics and requirements of each application.

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Data Availability

Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

Conflicts of Interest

The authors declare no conflict of interest.

References

- Agatz N, Bouman P, Schmidt M (2018). Optimization approaches for the traveling salesman problem with drone. *Transportation Science* 52(4): 965-981.
- Alfandari L, Ivana L, Marcos D M (2022). A tailored Benders decomposition approach for last-mile delivery with autonomous robots. *European Journal of Operational Research* 299: 510-525.
- Bai X, Ye Y, Zhang B, Ge S S (2023). Efficient package delivery task assignment for truck and high capacity drone. *IEEE Transactions on Intelligent Transportation Systems* 17(1): 248-260.
- Boccia M, Masone A, Sforza A, Sterle C (2021). A column-and-row generation approach for the flying sidekick travelling salesman problem. *Transportation Research Part C* 124(4): 102913.
- Boysen N, Schwerdfeger S, Weidinger F (2018). Scheduling last-mile deliveries with truck-based autonomous robots. *European Journal of Operational Research* 271: 1085-1099.
- Cai Y J, Choi T M (2021). Extended producer responsibility: A systematic review and innovative proposals for improving sustainability. *IEEE Transactions on Engineering Management* 68(1): 272-288.
- Cavani S, Iori M, Roberti R (2021). Exact methods for the traveling salesman problem with multiple drones. *Transportation Research Part C: Emerging Technologies* 130: 103280.

- Chang Y S, Lee H J (2018). Optimal delivery routing with wider drone-delivery areas along a shorter truck-route. *Expert Systems with Applications* 104: 307-317.
- Chen C, Demir E, Huang Y (2021a). An adaptive large neighborhood search heuristic for the vehicle routing problem with time windows and delivery robots. *European Journal of Operational Research* 294(3): 1164-1180.
- Chen C, Demir E, Huang Y, Qiu R (2021b). The adoption of self-driving delivery robots in last mile logistics. *Transportation Research Part E* 146: 102214.
- Chiang W C, Li Y, Shang J, Urban T L (2019). Impact of drone delivery on sustainability and cost: Realizing the UAV potential through vehicle routing optimization. *Applied Energy* 242: 1164-1175.
- Chung S H, Sah B, Lee J (2020). Optimization for drone and drone-truck combined operations: A review of the state of the art and future directions. *Computers & Operations Research* 123: 105004.
- Dayarian I, Savelsbergh M, Clarke J P (2020). Same-day delivery with drone resupply. *Transportation Science* 54(1): 229-249.
- Dell'Amico M, Montemanni R, Novellani S (2019). Drone-assisted deliveries: New formulations for the flying sidekick traveling salesman problem. *Optimization Letters* 15: 1-32.
- Dell'Amico M, Montemanni R, Novellani S (2021). Algorithms based on branch and bound for the flying sidekick traveling salesman problem. *Omega* 104(4): 102493.
- Dienstknecht M, Boysen N, Briskorn D (2022). The traveling salesman problem with drone resupply. *OR Spectrum* 44(4): 1045-1086.
- Dukkanci O, Koberstein A, Kara B Y (2023). Drones for relief logistics under uncertainty after an earthquake. *European Journal of Operational Research* 310: 117-132.
- El-Adle A M, Ghoniem A, Haouari M (2023). A variable neighborhood search for parcel delivery by vehicle with drone cycles. *Computers and Operations Research* 159: 106319.
- Gonzalez-R P L, Canca D, Andrade-Pineda J L, Calle M, Leon-Blanco J M (2020). Truck-drone team logistics: A heuristic approach to multi-drop route planning. *Transportation Research Part C: Emerging Technologies* 114: 657-680.
- Gu R, Poon M, Luo Z, Liu Y, Liu Z (2022). A hierarchical solution evaluation method and a hybrid algorithm for the vehicle routing problem with drones and multiple visits. *Transportation Research Part C* 141: 103733.
- Ha Q M, Deville Y, Pham Q D, Minh H H (2018). On the min-cost traveling salesman problem with drone. *Transportation Research Part C* 86: 597-621.
- Heimfarth A, Ostermeier M, Alexander H (2022). A mixed truck and robot delivery approach for the daily supply of customers. *European Journal of Operational Research* 303: 401-421.
- Imran N M, Mishra S, Won M (2023). A-VRPD: Automating drone-based last-mile delivery using self-driving cars. *IEEE Transactions on Intelligent Transportation Systems* 24(9): 9599-9612.
- Jacobsen S K, Madsen O B G (1980). A comparative study of heuristics for a two-level routing-location problem. *European Journal of Operational Research* 5(6): 378-387.
- Jeong H Y, Song B D, Lee S (2019). Truck-drone hybrid delivery routing: Payload-energy dependency and no-fly zones. *International Journal of Production Economics* 214: 220-233.
- Kang M, Lee C (2021). An exact algorithm for heterogeneous drone-truck routing problem. *Transportation Science* 55(5): 1088-1112.
- Karak A, Abdelghany K (2019). The hybrid vehicle-drone routing problem for pick-up and delivery services. *Transportation Research Part C: Emerging Technologies* 102: 427-449.
- Kitjacharoenchai P, Min B C, Lee S (2020). Two echelon vehicle routing problem with drones in last mile delivery. *International Journal of Production Economics* 225: 107598.
- Kitjacharoenchai P, Ventresca M, Moshref-Javadi M, Lee S, Tanchoco J M A, Brunese P A (2019). Multiple traveling salesman problem with drones: Mathematical model and heuristic approach. *Computers & Industrial Engineering* 129: 14-30.
- Kloster K, Moeini M, Vigo D, Wendt O (2023). The multiple traveling salesman problem in presence of drone and robot-supported packet stations. *European Journal of Operational Research* 305(2): 630-643.
- Kuo R J, Edbert E, Zulvia F E, Lu S H (2023). Applying NSGA-II to vehicle routing problem with drones considering makespan and carbon emission. *Expert Systems with Applications* 221: 119777.
- Lei D, Cui Z, Li M (2022). A dynamical artificial bee colony for vehicle routing problem with drones. *Engineering Applications of Artificial Intelligence* 107: 104510.

- Li H, Chen J, Wang F, Bai M (2021). Ground-vehicle and unmanned-aerial-vehicle routing problems from two-echelon scheme perspective: A review. *European Journal of Operational Research* 294(3): 1078-1095.
- Liang Y J, Luo Z X (2022). A survey of truck-drone routing problem: Literature review and research prospects. *Journal of the Operations Research Society of China* 10(2): 343-377.
- Liu D, Yan P, Pu Z, et al. (2021). Hybrid artificial immune algorithm for optimizing a van-robot e-grocery delivery system. *Transportation Research Part E: Logistics and Transportation Review* 154: 102466.
- Liu Y, Liu Z, Shi J, Wu G, Pedrycz W (2020). Two-echelon routing problem for parcel delivery by cooperated truck and drone. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 51(12): 7450-7465.
- Lu Y, Yang C, Yang J (2022). A multi-objective humanitarian pickup and delivery vehicle routing problem with drones. *Annals of Operations Research* 319: 291-353.
- Lu Y, Yang J, Yang C (2023). A humanitarian vehicle routing problem synchronized with drones in time-varying weather conditions. *Computers & Industrial Engineering* 184: 109563.
- Luo Q, Wu G, Ji B, Wang L, Suganthan P N (2021). Hybrid multi-objective optimization approach with Pareto local search for collaborative truck-drone routing problems considering flexible time windows. *IEEE Transactions on Intelligent Transportation Systems* 23(8): 13011-13025.
- Macrina G, Pugliese L D P, Guerriero F, Laporte G (2020). Drone-aided routing: A literature review. *Transportation Research Part C: Emerging Technologies* 120: 102762.
- Madani B, Ndiaye M (2022). Hybrid truck-drone delivery systems: A systematic literature review. *IEEE Access* 10: 92854-92878.
- Mara S T W, Achmad P R, Bertha M S (2022). An adaptive large neighborhood search heuristic for the flying sidekick traveling salesman problem with multiple drops. *Expert Systems with Applications* 205: 117647.
- Masmoudi M A, Mancini S, Baldacci R, Kuo Y H (2022). Vehicle routing problems with drones equipped with multi-package payload compartments. *Transportation Research Part E: Logistics and Transportation Review* 164: 102757.
- Meng S, Guo X, Li D, Liu G (2023). The multi-visit drone routing problem for pickup and delivery services. *Transportation Research Part E: Logistics and Transportation Review* 169: 102990.
- Momeni M, Mirzapour Al-e-Hashem S M J, Heidari A (2023). A new truck-drone routing problem for parcel delivery by considering energy consumption and altitude. *Annals of Operations Research* 337: 1-47.
- Moshref-Javadi M, Hemmati A, Winkenbach M (2020a). A truck and drones model for last-mile delivery: A mathematical model and heuristic approach. *Applied Mathematical Modelling* 80: 290-318.
- Moshref-Javadi M, Lee S, Winkenbach M (2020b). Design and evaluation of a multi-trip delivery model with truck and drones. *Transportation Research Part E* 136: 101887.
- Moshref-Javadi M, Winkenbach M (2021). Applications and research avenues for drone-based models in logistics: A classification and review. *Expert Systems with Applications* 177: 114854.
- Murray C C, Chu G A (2015). The flying sidekick traveling salesman problem: Optimization of drone-assisted parcel delivery. *Transportation Research Part C* 54: 54-109.
- Murray C C, Raj R (2020). The multiple flying sidekicks traveling salesman problem: Parcel delivery with multiple drones. *Transportation Research Part C* 110: 268-398.
- Najy W, Archetti C, Diabat A (2023). Collaborative truck-and-drone delivery for inventory-routing problems. *Transportation Research Part C: Emerging Technologies* 146: 103791.
- Ostermeier M, Heimfarth A, Alexander H (2023). The multi-vehicle truck-and-robot routing problem for last-mile delivery. *European Journal of Operational Research* 310: 680-697.
- Pina-Pardo J C, Silva D F, Smith A E (2021). The traveling salesman problem with release dates and drone resupply. *Computers & Operations Research* 129: 105170.
- Poikonen S, Golden B (2020). Multi-visit drone routing problem. *Computers and Operations Research* 113: 104802.
- Raj R, Murray C C (2020). The multiple flying sidekicks traveling salesman problem with variable drone speeds. *Transportation Research Part C* 120: 102813.
- Rave A, Fontaine P, Kuhn H (2023). Drone location and vehicle fleet planning with trucks and aerial drones. *European Journal of Operational Research* 308(1): 113-130.
- Roberti R, Ruthmair M (2021). Exact methods for the traveling salesman problem with drone. *Transportation Science* 55(2): 315-335.

- Sacramento D, Pisinger D, Ropke S (2019). An adaptive large neighborhood search metaheuristic for the vehicle routing problem with drones. *Transportation Research* 102: 289-315.
- Salama M R, Srinivas S (2022). Collaborative truck multi-drone routing and scheduling problem: Package delivery with flexible launch and recovery sites. *Transportation Research Part E* 164: 102788.
- Schermer D, Moeini M, Wendt O (2019a). A hybrid VNS/Tabu search algorithm for solving the vehicle routing problem with drones and en route operations. *Computers & Operations Research* 109: 134-158.
- Schermer D, Moeini M, Wendt O (2019b). A matheuristic for the vehicle routing problem with drones and its variants. *Transportation Research Part C: Emerging Technologies* 106: 166-204.
- Simoni M D, Kutanoglu E, Claudel C G (2020). Optimization and analysis of a robot-assisted last mile delivery system. *Transportation Research Part E: Logistics and Transportation Review* 142: 102049.
- Sluijk N, Florio M A, Kinable J, Dellaert N, Woensel V T (2023). Two-echelon vehicle routing problems: A literature review. *European Journal of Operational Research* 304: 865-886.
- Srinivas S, Ramachandiran S, Rajendran S (2022). Autonomous robot-driven deliveries: A review of recent developments and future directions. *Transportation Research Part E* 165: 102834.
- Tamke F, Buscher U (2021). A branch-and-cut algorithm for the vehicle routing problem with drones. *Transportation Research Part B* 144: 174-203.
- Thomas T, Srinivas S, Rajendran C (2023). Collaborative truck multi-drone delivery system considering drone scheduling and en route operations. *Annals of Operations Research* 339: 1-47.
- Tinic G O, Karasan O E, Kara B Y, Campbell J F, Ozel A (2023). Exact solution approaches for the minimum total cost traveling salesman problem with multiple drones. *Transportation Research Part B: Methodological* 168: 81-123.
- Wang K, Yuan B, Zhao M, Lu Y (2020). Cooperative route planning for the drone and truck in delivery services: A bi-objective optimisation approach. *Journal of the Operational Research Society* 71(10): 1657-1674.
- Wang Y, Wang Z, Hu X, Xue G, Guan X (2022). Truck-drone hybrid routing problem with time-dependent road travel time. *Transportation Research Part C* 144: 103901.
- Wang Z, Sheu J B (2019). Vehicle routing problem with drones. *Transportation Research Part B: Methodological* 122: 350-364.
- Wu G, Mao N, Luo Q, Xu B, Shi J, Suganthan P N (2022). Collaborative truck-drone routing for contactless parcel delivery during the epidemic. *IEEE Transactions on Intelligent Transportation Systems* 23(12): 25077-25091.
- Xia Y, Zeng W, Zhang C, Yang H (2023). A branch-and-price-and-cut algorithm for the vehicle routing problem with load-dependent drones. *Transportation Research Part B: Methodological* 171: 80-110.
- Yang Y, Yan C, Cao Y, Roberti R (2023). Planning robust drone-truck delivery routes under road traffic uncertainty. *European Journal of Operational Research* 309(3): 1145-1160.
- Yin Y, Li D, Wang D, Ignatius J, Cheng T C E, Wang S (2023a). A branch-and-price-and-cut algorithm for the truck-based drone delivery routing problem with time windows. *European Journal of Operational Research* 309(3): 1125-1144.
- Yin Y, Yang Y, Yu Y, Wang D, Cheng T C E (2023b). Robust vehicle routing with drones under uncertain demands and truck travel times in humanitarian logistics. *Transportation Research Part B* 174: 102781.
- Yu S, Puchinger J, Sun S (2022). Van-based robot hybrid pickup and delivery routing problem. *European Journal of Operational Research* 298: 894-914.
- Zhang G, Zhu N, Ma S, Xia J (2021). Humanitarian relief network assessment using collaborative truck-and-drone system. *Transportation Research Part E* 152: 102417.

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